

Practice set 2

Exponentials and logarithms

Write in the spaces

Answers · official assessment schedule extracts on the last pages

- (c) Jessica is investigating a compounding investment. She wants to know how long it would take for an investment of \$1000 to double in value to \$2000. She forms the following equation:

$$2000 = 1000 \left(1 + \frac{R}{100} \right)^D$$

where R is the rate of return on the investment, as a percentage, and

D is the time that the investment would take to double in value, in years.

- (i) If an investment takes 11 years to double in value, what is its rate of return?

- (ii) By making D the subject of the expression: $2000 = 1000 \left(1 + \frac{R}{100} \right)^D$,

show that $D = \frac{\log(2)}{\log(1 + \frac{R}{100})}$

In her research, Jessica comes across a simple but **approximate** rule for calculating D , the time that the investment would take to double in value. It is commonly called the ‘Rule of 72’, and it states that:

$$D = \frac{72}{R}$$

Jessica wonders how close the values of D from the ‘Rule of 72’ are to those calculated using the actual expression, which is:

$$D = \frac{\log(2)}{\log(1 + \frac{R}{100})}$$

- (iii) Show clearly that the value of R for which the ‘Rule of 72’ exactly calculates D , is the solution to the equation:

$$2^R - \left(1 + \frac{R}{100}\right)^{72} = 0$$

You do not need to solve this equation.

(ii) Solve the equation $2^x = 2022$.

(b) Simplify the following expression fully, writing your answer as a single logarithm.

$$\log(3a) + 2\log\left(\frac{a}{6}\right)$$

- Explaining your reasoning clearly, describe which values of a and of c will make the equation possible to solve.

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Q TWO	Expected coverage	Achievement (u)	Merit (r)	Excellence (t)
(c)(i)	$D = 11$, so $\left(1 + \frac{R}{100}\right)^{11} = 2$ $\left(1 + \frac{R}{100}\right) = \sqrt[11]{2} = 1.065$ $R = 6.5(\%)$	Sets up equation and uses the 11 th root. OR CAO	Obtains correct solution.	
(c)(ii)	$\left(1 + \frac{R}{100}\right)^D = 2$ $\log\left[\left(1 + \frac{R}{100}\right)^D\right] = \log 2$ $D \log\left(1 + \frac{R}{100}\right) = \log 2$ $D = \frac{\log(2)}{\log\left(1 + \frac{R}{100}\right)}$	Logs taken of both sides (one of lines 2 or 3 must be shown).	Given expression derived correctly (one of lines 2 or 3 must be shown).	
(c)(iii)	$\frac{72}{R} = \frac{\log(2)}{\log\left(1 + \frac{R}{100}\right)}$ $72 \log\left(1 + \frac{R}{100}\right) = R \log(2)$ $\log\left(\left(1 + \frac{R}{100}\right)^{72}\right) = \log(2^R)$ $\left(1 + \frac{R}{100}\right)^{72} = 2^R$ $2^R - \left(1 + \frac{R}{100}\right)^{72} = 0$	Sets equations equal.	Processes powers (line 3)	T1: Obtains given equation with incorrect mathematical statements. T2: Obtains final given equation with correct mathematical statements. <i>PL: It may be reqd to make T1 = (c)(ii), T2=(c)(iii)</i>

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Q	Evidence	Achievement	Merit	Excellence
(ii)	$x \log(2) = \log(2022)$ $x = 10.98$ Accept $\log_2(2022)$	Correct solution.		
(b)	$\log(3a) + 2 \log\left(\frac{a}{6}\right)$ $= \log(3a) + \log\left(\left(\frac{a}{6}\right)^2\right)$ $= \log\left(3a \left(\frac{a}{6}\right)^2\right)$ $= \log\left(\frac{a^3}{12}\right)$	Fraction not correctly simplified but otherwise correct.	Correct expression obtained with fraction correctly simplified.	

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Q	Evidence	Achievement	Merit	Excellence
(c)(i)	$\log_2(x-a) - \log_2(x+a) = c$ $\log_2 \frac{x-a}{x+a} = c$ $\frac{x-a}{x+a} = 2^c$ $x-a = 2^c(x+a) = x2^c + a2^c$ $x(1-2^c) = a + a2^c = a(1+2^c)$ so, $x = a \frac{1+2^c}{1-2^c}$	Log expressions combined correctly.	Correct exponential equation obtained (line 3).	Correct mathematical statements lead to the required expression.

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Q	Evidence	Achievement	Merit	Excellence
(ii)	Using the expression from (c) part (i) ... Firstly, if x is not defined, there will be no solutions, so that means that $1 - 2^c \neq 0$, so $2^c \neq 1$, and $c \neq 0$. Hence c cannot be zero. Secondly, if $a = 0$, then $x = 0$, but then the logs will be undefined. Hence, a cannot be zero. [Although, in the original equation, if $a = 0$ and $c = 0$, any strictly positive x -value is a solution, but the expression for x is undefined] Thirdly, for the original equation to be defined, both $x - a > 0$ and $x + a > 0$ (accept one or the other, or both).		One constraint identified with reasoning.	Two constraints identified with reasoning.